

Kaggriculture: what we tried

A complete record of every method attempted, the measured result, and the peak performance reached. Competition runs 2026-07-29 to 2026-09-30. Report generated 2026-08-24.

Highest value achieved

Measure	Value	Where
Best skill rating	1747.4	submission 55563850 (lookahead 3), now retired
Active ratings	1698.9 and 1667.1	55677927 and 55657021, both converged
Best leaderboard rank	~#1,069 of 6,046	top 18% of the field
Highest single-game bank	\$172,590	vs pwdh2026, won by \$99,884
Largest winning margin	\$139,539	vs Kaushik, banked \$139,660
Mean bank across 486 live games	\$90,696	median \$88,774
Best live win rate	54.6%	55563850 over 269 games

The rating is the real score: the ladder counts only win, loss or tie, and the coin margin is discarded. A \$172,590 bank and a \$325 win move the rating identically.

Methods that worked (all shipped, all re-validated)

Each edge was re-tested by playing the agent against the version of itself without that edge, live, both seats, on two disjoint blocks of seeds it had never been screened on. Combined records below.

Method	What it does	Held-out record	Earlier estimate
Preemption lookahead 3	Sells a premium line up to 3 steps before the schedule's own sale	58-2	31/32, SPRT 7-0-1
Eager seller	Sells EGG, CARROT and TOMATO off the real price ladder while a unit still fetches 0.3x base	43-5	+51 wins
Carrot swap	Grows the 13 end-of-season wheat plantings as carrot instead	40-8	+12 wins
Carrot price gate	At day 26 picks carrot or wheat from the actual market, all 8 shops known	25-11	+3 wins
Weed repair	Substitutes DIG when a weed blocks a scheduled PLANT, then resyncs	inherited	in all top agents

Why these worked and nothing else did: each one **adds output or sales using turns already spent**, or substitutes an action the route already performs. The carrot swap is the clearest case -- carrot waters at ages 2-3 and the route's end-of-season wheat is lifted at age 3, so carrot fits exactly the hole wheat leaves. It was never that carrot is a good crop.

Methods that did not work

Production and route changes

Method	Result	Why it failed
Melon patch (extra melon tiles)	-99 wins	Route carries choreography only for the assets it expects
Goose herd swap	-105 wins	Same; eggs also glut on 2 shops at a \$50 base
Day-0 cow-for-sheep swap	-\$42k	Ends with 5 empty pastures; PLACE needs a matching structure
Pasture retargeting	-2 wins	No effect worth the risk
Tomato swap (4 donor sets)	-146 to -190 wins	An <i>ongoing</i> crop yields max_yield units in TOTAL then dies -- tomato is 4 units at interval 1, worth half a strawberry tile
Fertilize wheat, age 2	-3,335 margin	Two dry days turn a plant to WEED; wheat's window opens at age 2 so removing that watering kills it
Fertilize wheat, age 3	-96 (neutral)	Arithmetically identical to not fertilizing
Fourth quadrant (SE land)	10-0 loss	Tested by Rayk Kretzschmar over 450 games; we did not repeat it
Role specialisation (CMA-ES)	no effect	Measured, left off; the +\$8,399 was an RNG artifact worth \$415

Market and timing changes

Method	Result	Why it failed
Feed-buy ceiling (30 / 20 / 12)	-44 / -49 / -105 wins	The shed's wheat feeds the herd AND scheduled sales; at 12 the animals starve
Sale meter, price floor	-26 / -32 / -47 wins	A price floor is not a rate limit: admits 32-40 units at equilibrium, 0 once glutted, so it empties the shed early then goes inert
Sale meter, town rate cap	-51 / -64 / -75 wins	Milk, strawberry and wool CLEAR the market. Selling earlier only raises average inventory, and average inventory sets average price
Eager-sell fertilizer	-151 wins	The route fertilizes 100% of its strawberry from that same shed
Eager-sell fertilizer + wheat	-208 wins	Gives back every win: shed wheat is the feed chain
Front-running / SELL reordering	0 coins available	Both we and rivals already sell in slot ~0.3; the race is a dead heat

Whole-agent replacements

Method	Result	Note
CMA-ES policy agent	rating 841-892	vs 1747 for the route agent
Lifted route v15 (iVl44d)	6/16	\$12k worse than the route it beat
Lifted ReCurSiON (#17)	0/6	-4,442 a game
Lifted Ryo Hasegawa (#1)	0/6	-6,486. Rank does not predict transplantability
Lifted Arman Tuganbaev (#4)	0/6	-11,098 a game
Lifted MiMi (#5)	6/6, then 8-32	Won on the 3 seeds it was picked with, lost badly on 20 held-out seeds

What the failures taught (the transferable part)

Rule	Evidence
Markets that clear cannot be gamed on timing. Milk, strawberry and wool finish within ~40 units of equilibrium. When supply and absorption are both fixed, selling earlier only raises average inventory and lowers average price.	Sale meter lost at every rate; turning it on raised mean market inventory and dropped our bank 84,230 to 79,879 with the opponent's unchanged
Selling early is free only where demand is never met. Carrot ends -307, egg -230, tomato -225. Those are exactly what the eager seller targets.	Eager seller +51 then 43-5; carrot swap 40-8
A crop substitution works only where the donor's existing actions fit the new crop's calendar.	Carrot fits wheat's age-3 lift and pays. Tomato, melon, geese all needed actions the recording does not contain: 0 for 8
An item the farm consumes is not surplus.	Fertilizer sells for 13 and still is not spare; wheat sells and still is the feed
Our replay benchmark detects breakage but cannot rank comparable agents.	It scored MiMi's route +24 wins over the incumbent that then beat it 32-8. Wrong sign, not noise
Decide on seeds the candidate has never seen.	MiMi went 6/6 on its screening seeds and 8-32 held out

Current state

Item	Value
Active submissions	55677927 (1698.9) and 55657021 (1667.1)
Shipped configuration	LOOKAHEAD 3, EAGER_FLOOR_FRAC 0.3, CARROT_SWAP 13 plantings, CARROT_PRICE_GATE 1.0
Disabled and kept as measured negatives	METER_RATE, TOMATO_SWAP, FERTILIZE_WHEAT_AGE, HERD_SWAP, PASTURE_TILT, MELON_PATCH, FEED_BUY_CEILING
Slot correlation	100% -- the two submissions win and lose the identical games, so the second slot currently provides no hedge
Final ranking rule	Only the best-scoring bot is shown, so a weaker second submission cannot drag the team down
Remaining timeline	Final submission 2026-09-30; games continue ~2 weeks; board final ~2026-10-15

Tally: 5 methods shipped and validated, 21 measured and rejected. The rejected work is the larger half of the record and the more useful one -- four of the six rules above came from failures, and the benchmark-validity finding came from the single false positive.